Anexo I

UNIVERSIDADE DE ÉVORA

Cloud Computing

1º Trabalho

2025/2026

Sponsor a Donkey platform 1.0

Objective
To help local producers and reduce the risk of extinction of the Burro de Miranda species, an association in northern Portugal wants a digital platform to raise awareness of these animals, while simultaneously gathering international support. This would involve creating connections between citizens who make donations over time (the sponsor) and the (sponsored) animals, whose daily life is documented in posts (photo and text) for their followers.
Students will design, estimate costs, and deploy a scalable and cost-effective solution for this case.

Phase 1: Design
Students will design the architecture of the web application using GCP services.

Functional Requirements: the solution should include a web interface for:

- Registered users with a producer profile
- Registered users with a sponsor profile
- Unregistered users who, upon seeing the activity, may wish to register
- Information about a donkey (initial registration; recent activity record – example: “spent the day running” and a photo of it)
- Generation of a “sponsor news” message, compiling the most recent posts about the donkey, to be sent (only in a simulated way or via internal e-mail using an Apps Script service).

Non-Functional Requirements:

- Scalability and Cost-effectiveness.

Phase 2: Cost Estimation
Produce a report detailing the estimated costs, including breakdowns by service or component, and optionally with multiple scenarios.

Phase 3: Deployment
Develop a basic web application (e.g., using a framework like Flask, Node.js or Java Spring Web MVC).
Students will deploy their solution to the chosen GCP service, within a project (suggested project name m99888ccproj01, where 99888 is the student number).
This project and all used resources must be shared (for reading, but not for writing), through the procedure you deem appropriate (explain in the report) with the UC teacher.

Any omitted aspects are decided by the students and mentioned in the report.
The project will be presented at the end.

If there is difficulty in some phase, the student may skip and simulate that part, and perform the remaining parts.

Submission
The work must be completed within the established deadline, ending with the Moodle submission of:

- the web application source code;
- report with (group) students’ identification, and work remarks (without exceeding 3 pages).

Students can work in groups of two or individually.

Deadline
Defined in Moodle